Know When to Fold: Distribution-Shift-Aware Selective Prediction for Protein–Ligand Co-Folding

MoML 2026 short paper (draft). Non-archival. Rikhin Kavuru.

Abstract

Protein–ligand co-folding models (AlphaFold3, Boltz, Chai) report confidence scores that correlate with pose accuracy, but a correlation is not a decision rule, and the correlation is weakest exactly where drug discovery operates: on novel pockets and novel chemotypes. We recast co-folding reliability as **selective prediction** and build a model-agnostic, training-free layer that turns confidence into a risk-controlled accept/abstain decision with a finite-sample guarantee. On the released Runs N' Poses benchmark (five governed co-folding models, 12,602 delivered poses, plus a Boltz-2 affinity comparator for 13,535 in the raw table) we show three things. (1) A conformal selective-risk gate is valid on i.i.d. data. (2) That guarantee **breaks** under the novelty shift central to drug discovery: a gate whose marginal error looks compliant under-controls error roughly twofold on novel-ligand strata, and calibrating on familiar ligands then deploying on novel ones violates the guarantee in 100% of runs (realized error 0.55 against a 0.20 target). (3) The break is repaired by group-conditional and weighted conformal keyed on training-set similarity, and a combined reliability score cuts the area under the risk-coverage curve (AURC) by roughly a quarter to two-fifths over native confidence (paired data-bootstrap over test poses excludes zero for all five models tested; Boltz-2 omitted for power), roughly doubling to tripling the predictions retained at a fixed guarantee. Adding structural pose-agreement features from the model's own diffusion samples pushes the AURC reduction to 38–51% across models (pose Δ(AURC) CI excludes zero for all five), and abstention raises a genuinely downstream, non-circular metric, recovery of the correct crystal contacts (0.90 vs 0.72 on accepted vs rejected poses). The layer is released as a pip-installable package that wraps frozen model outputs.

1. Introduction

Co-folding models are increasingly used to prioritise molecules and generate starting structures for downstream design. Recent work establishes that their confidence tracks accuracy and that accuracy drops on inputs dissimilar to training (Runs N' Poses, FoldBench). Neither gives a practitioner what they need: a rule for **which predictions to trust**, with a guarantee that survives the distribution shift they actually face.

We provide that rule. Our contribution is (i) selective prediction / conformal risk control for co-folding **pose** reliability, a structured 3D output rather than a scalar property. The closest prior work, CoDrug, conformalizes a scalar molecular property under shift, and we are not aware of a prior conformal treatment of pose correctness. Further, (ii) a sharp, quantified demonstration that standard conformal guarantees collapse under novel-pocket / novel-chemotype shift, using training-set structural and chemical similarity as the shift variable; and (iii) a shift-robust repair (group-conditional and importance-weighted conformal keyed on that similarity) plus a combined reliability score, packaged as a training-free layer over frozen AlphaFold3 (AF3) / Boltz / Chai outputs. We show ordinary probability calibration is not a substitute: it carries no finite-sample coverage and breaks under the same shift, whereas the conformal variants repair it.

2. Method

Setup. The unit is the delivered pose: the top-1 sample per (system, ligand, model) by the model's own ranking score. The label is Y = 1 iff the symmetry-corrected ligand root-mean-square deviation (RMSD) is $\leq$ 2 Å. The gate accepts a pose when its confidence s $\geq \tau$ and abstains otherwise; we report selective risk (error among accepted) against coverage (fraction accepted).

Certified threshold. We choose τ by Learn-then-Test with fixed-sequence testing [Angelopoulos et al. 2021] over a pre-specified, data-independent sequence of top-k accept ranks (smallest first). Each null H0(τ): P(error | s $\geq \tau$) $\geq \alpha$ is tested with the binomial tail p-value P(Bin(n_accept, α) $\leq$ errors); the accept set grows while H0 is rejected at level δ and stops at the first failure, giving P(selective risk $\leq \alpha$) $\geq$ 1 -

δ, finite-sample and distribution-free. The p-value is exact for a homogeneous error rate and conservative (hence still valid) for the heterogeneous Poisson-binomial case by a convex-order argument; conditioning on the accept count is the correct tool for error-among-accepted.

Novelty and shift-robustness. Training-set similarity (ECFP4 Tanimoto to the nearest training ligand, pocket similarity, temporal cutoff), shipped pre-computed by Runs N' Poses, is the covariate-shift variable. It both stratifies (group-conditional / Mondrian conformal: a separate τ per novelty stratum) and weights (weighted conformal [Tibshirani et al. 2019]: reweight familiar-source calibration points by an estimated likelihood ratio to certify on a novel target without target labels). Ligands with no training analog form their own extreme stratum.

Combined score. Native ranking score is a weak sorter for some models. We fit a calibration-only combiner (gradient boosting $\rightarrow$ P(correct)) over native confidence, interface chain-pair ipTM, PoseBusters physical validity, intra-model ensemble spread across diffusion samples, ligand difficulty, and cross-model agreement where other models' predictions are available (a single-model deployment drops this feature, keeping the 22–38% gain without it). "Training-free" here means the co-folding model is never retrained; the combiner is a small model fit only on the calibration fold, so the layer is free of structure-predictor training, not of all fitting. Novelty is excluded from the score and enters only through calibration. A 3-way split (fit combiner / calibrate τ / test) preserves conformal validity for the learned score: the combiner never sees the test fold and the threshold is calibrated on out-of-sample combiner scores, so the learned score carries no combiner-fit leakage (target-clustering across folds is addressed separately under Utility).

3. Results

Data: released Runs N' Poses predictions consumed as-is (no inference), five governed models (12,602 delivered poses); a Boltz-2 affinity head brings the raw table to 13,535 and is used only as an ungoverned comparator. $\alpha = 0.20$, $\delta = 0.10$ unless noted.

Raw novelty gradient. AF3 delivered-pose correctness falls from 0.88 on familiar ligands to 0.44 on the most novel with-analog stratum. A single global threshold cannot hold a uniform guarantee across this gradient.

Valid i.i.d. The certifier's finite-sample guarantee, true selective risk $\leq \alpha$ with probability 1 - δ, is validated on synthetic data where the true risk is known (the gate holds in $\geq$ 1 - δ of draws). The realized-risk indicator on a finite test fold is a *downward-biased* proxy for that event, and we demonstrate the bias rather than assert it: on synthetic data the small-fold indicator dips below 1 - δ while the large-sample (true-risk) indicator meets it. On RNP the fraction of splits holding, with a Clopper-Pearson interval, reaches the 1 - δ = 0.90 target for Boltz-1x (0.90 [0.88, 0.93]) and approaches it for Boltz-1 (0.88 [0.85, 0.91]) and AF3 (0.85 [0.81, 0.89]), whose interval sits just below 0.90; the tight-certifier realized-risk indicator is the reason it is a downward-biased proxy rather than the guarantee itself, and the certified gate's mean realized risk is $\leq \alpha$ there (AF3 0.17 at 28% coverage). Native ranking score certifies almost nothing for Chai and Protenix ($<$ 5% coverage): a near-vacuous gate that motivates the combined score below.

The break. AF3's marginal risk (0.177) hides severe per-stratum under-control:

novelty stratum	S0	S1	S2	S3	S4
realized selective risk	0.07	0.15	0.16	**0.38**	**0.43**

Calibrating on familiar ligands and deploying on novel ones accepts 98% of poses at realized error **0.55** with the guarantee holding in **0%** of runs (violated in 100%). All five governed models show the same collapse. The break is a property of structural/chemical novelty, not recency: it is at least as strong along pocket-novelty (AF3 S0 0.06 $\rightarrow$ S3 0.40, Chai up to 0.77) but weak along a temporal release-date axis, which sharpens what the shift variable must capture.

The repair. Group-conditional calibration (a separate τ per novelty stratum) restores per-stratum validity: each stratum's accepted error is $\leq \alpha$ with probability 1 - δ marginally (a simultaneous across-strata statement follows by calibrating each at δ/S over the S strata). With native confidence the cost is heavy abstention on the hardest strata (the gate folds rather than assure falsely). With the combined score, group-conditional calibration instead recovers usable coverage across the gradient while holding risk $\leq \alpha$: AF3 accepts 52% of stratum S1 and 39% of S2 (vs 8% and 12% with native confidence) and certifies a small fraction of the novel S3 that native confidence must abstain on. Only the no-training-analog extreme (S4) stays uncertifiable, where abstention is the correct answer.

Utility. The combined score dominates native confidence on the risk-coverage curve:

model	AURC native	AURC combined	Δ
AF3	0.187	0.125	-33.1%
Boltz-1	0.234	0.170	-27.5%
Boltz-1x	0.216	0.163	-24.6%
Chai-1	0.248	0.150	-39.6%
Protenix	0.281	0.175	-37.7%

The AURC reduction is significant per model: a paired bootstrap over test poses gives 90% CIs on Δ(AURC) that exclude zero (AF3 0.070 [0.049, 0.092], Protenix 0.092 [0.068, 0.116]; Boltz-2 omitted for power, n = 933). Because a target recurs across ~5 poses, the pooled random split leaks 95% of test targets into calibration; a target-grouped split and a target-cluster bootstrap give the same conclusion (AF3 Δ(AURC) [0.056, 0.079]), so the gain is not an artifact of clustering. At $\alpha = 0.2$ the combined gate retains far more predictions at the same guarantee (AF3 coverage 0.22 → 0.71; Chai 0.01 → 0.60) and unlocks the stringent $\alpha = 0.1$ (90%-correct) operating point that native confidence cannot certify at all for several models. The combined score fuses native confidence with interface ipTM, PoseBusters validity, intra-model ensemble spread, and cross-model agreement.

Weighted repair, label-free. Calibrating on familiar ligands and correcting with cross-fitted, probability-calibrated likelihood-ratio weights over the novelty covariates pulls the realized error on a moderately-novel target toward α without target labels (AF3 naive 0.27 → weighted 0.19 at 0.70 coverage; same trend for all models, and robust to the weight-model choice). We also implement an importance-weighted Learn-then-Test gate [Almeida et al. 2025] built on a WSR betting p-value [Waudby-Smith & Ramdas 2024] that is finite-sample valid *conditional on correct weights*; on real co-folding data it abstains, because even the plug-in barely clears α, so no certifiable margin remains. A concept-shift diagnostic explains why: P(correct | confidence) itself moves between source and target, and the gap grows from the moderate regime (0.08–0.16) to the extreme regime (0.17–0.29), so pure covariate reweighting cannot restore validity on novel pockets. Weighted conformal is therefore the label-free complement; group-conditional calibration is the rigorous finite-sample guarantee where stratum labels exist. On the extreme regime (< 55% correct at baseline) weighted conformal reduces error (0.50 → 0.35) but no gate can certify a high-coverage low-error set, so the layer correctly abstains.

Baselines: calibration is not conformal. The combined score has the lowest AURC among the confidences a practitioner would reach for (AF3 0.12 vs native ipTM 0.16 vs ranking score 0.19; same ordering for all models), and PoseBusters validity alone is a poor gate (realized error 0.26–0.34). The decisive comparison isolates the guarantee from the features: the same combined score as a Platt/isotonic-calibrated classifier with a fixed threshold versus inside the conformal layer. i.i.d., both control error; **under the novelty shift the Platt-calibrated fixed-threshold gate breaks like naive conformal** (realized error 0.22 above α). Isotonic calibration sits near α (0.18) only by abstaining down to 71% coverage, and even then holds the guarantee in 76% of runs rather than the required 90%, since calibration carries no finite-sample coverage. Only the shift-robust conformal repair (group-conditional, which has no calibration analogue) restores realized error $\leq \alpha$ with a guarantee, by abstaining more on the novel strata.

Downstream payoff. The gate cleans the delivered pose set a downstream pipeline would carry forward: base top-1 purity 63–70% rises to 82% (α=0.2, retaining 86% of correct AF3 poses) or 91–94% (α=0.1, all models), with interface LDDT-PLI lifting from ~0.72 to 0.85–0.92.

A non-circular downstream payoff. Purity equals 1 - selective risk by construction, so E6 cannot show the gate buys anything beyond the guarantee. We add a genuinely different, downstream metric: protein-ligand *interaction-fingerprint recovery* (receptor residues within 4.5 Å of the ligand vs the crystal structure), which is only correlated with, not equal to, the 2 Å label. Under the gate, accepted poses recover 0.90–0.91 of the crystal contacts vs 0.71–0.76 for rejected poses, and the accepted-minus-rejected recall gap 90% CI excludes zero for all five models (gaps 0.14–0.21). Abstention routes forward the poses whose interaction patterns a downstream SAR or structure-based step can trust. A screening-enrichment harness (EF/BEDROC, selective-EF, coverage-enrichment and active-retention curves, random-abstention control) is implemented; the archival version applies it to a released co-folded screen (Shen et al.), where the governed pose signal enriches actives above docking but the headline enrichment rides on an ungoverned affinity ranker, so we scope that arm out here. The Mac1 prospective screen is the drop-in target when its coordinates release.

Robustness and ablation. The break and the gain do not depend on the 2 Å convention: across thresholds 1.5–3.0 Å the novel strata always exceed α (AF3 S3 risk 0.35–0.43 vs S0 0.05–0.09) and the combined score cuts AURC 31–34%. A cumulative feature ablation attributes the gain mainly to interface ipTM (AURC 0.19 $\rightarrow$ 0.16), intra-model ensemble spread (0.16 $\rightarrow$ 0.14), and cross-model agreement (0.14 $\rightarrow$ 0.13); PoseBusters and ligand difficulty add little on top.

Pose-agreement upgrade (structures). Beyond the released confidence tables, the predicted structures carry an orthogonal signal: whether the binding *modes* agree. From the model's own 25 diffusion samples we compute intra-model pose diversity, and across models the delivered-pose ligand-RMSD (spyrmsd, pocket-superposed, symmetry-corrected; over 11,711 poses, no GPU). Adding these lowers AURC further on top of every tabular feature (the single largest drop after ipTM, AF3 0.123 $\rightarrow$ 0.106), raising the overall reduction to 38–51% across models, the pose-only Δ(AURC) 90% CI excluding zero for all five. Intra-model pose diversity is a free byproduct of running the model, so this keeps the training-free framing; the tabular combined score remains the cheap, structure-free primary.

Task- and label-agnostic. The layer is not tied to the 2 Å pose label. On interface quality (local distance difference test on the protein-ligand interface, LDDT-PLI $\geq$ 0.5) the combined score again dominates native confidence on AURC (38–55% lower). Dropping the threshold entirely and ordering by continuous RMSD, the accepted set's mean RMSD at 50% coverage falls from ~2.1 Å (native) to ~1.1–1.5 Å (combined). We also certify a *continuous* gate (mean bounded-RMSD among accepted $\leq$ a target with probability 1 - δ) using a variance-adaptive WSR betting bound, which certifies non-trivial coverage where a distribution-free Hoeffding bound certifies almost none (AF3, target 1 Å mean-RMSD: WSR 43% coverage vs Hoeffding 0%).

4. Discussion and limitations

The guarantee is over the chosen correctness definition and the sampled shift axes, not a universal notion of trust; the findings are robust across RMSD thresholds 1.5–3.0 Å and extend to a certified continuous-RMSD gate. The finite-sample weighted-conformal guarantee is exact only conditional on correct importance weights, and under novel-pocket shift there is residual concept shift (P(correct | confidence) moves), so we keep group-conditional as the rigorous finite-sample guarantee and scope weighted conformal as the label-free complement. A cross-dataset test on FoldBench closes this once the missing feature is restored: the public per-pose table ships only ranking_score, so we regenerated the FoldBench Protenix predictions to recover interface-ipTM and self-scored ligand-RMSD against the deposited assemblies (436 targets, 384 train-similar, 52 low-homology; feature and label from one run). This compares gates, not benchmarks: our self-scored Protenix top-1 success is 0.40 against FoldBench's released 0.57, a gap from the scoring convention, model version, and MSA source that does not affect the transfer because feature and label are self-consistent. The frozen Runs N' Poses interface-ipTM gate then transfers with a risk-coverage AURC of 0.380, ahead of the matched ranking_score control at 0.454, and the same threshold accepts only 5% of FoldBench against 26% at home, so the coverage the gate can certify collapses on the low-homology second dataset. On the extreme novel-chemotype regime (base correctness < 55%) no gate can certify a high-coverage low-error set; the layer's value there is principled abstention over a naive gate's false confidence. Code, calibration tables, and a one-command reproduction pipeline are provided in the accompanying repository (public release on acceptance).

References

Abbreviated; full entries in the repository `REFERENCES.bib`. Angelopoulos, Bates, Candès, Jordan, Lei. Learn then Test. 2021. Bates, Angelopoulos, Lei, Malik, Jordan. Risk-Controlling Prediction Sets. JACM 2021. Tibshirani, Foygel Barber, Candès, Ramdas. Conformal Prediction Under Covariate Shift. NeurIPS 2019. Nguyen et al. CoDrug: Conformal Drug Property Prediction under Covariate Shift. NeurIPS 2023. Shen et al. Unlocking the application potential of AlphaFold3-like co-folding for virtual screening. Chem. Sci. 2025. Waudby-Smith, Ramdas. Estimating Means of Bounded Random Variables by Betting. JRSS-B 2024. Vovk. Conditional Validity of Inductive Conformal Predictors (Mondrian). Mach. Learn. 2013. Škrinjar et al. Have protein–ligand co-folding methods moved beyond memorisation? (Runs N' Poses) 2025. Buttenschoen, Morris, Deane. PoseBusters. Chem. Sci. 2024.